

## TEZPUR UNIVERSITY

End-Semester Examination, Autumn 2014

## CO 504 NATURAL LANGUAGE PROCESSING

Marks : 60

Time : 2 hours

*The figures in the right-hand margin indicate marks for the individual question.*

*Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.*

**Submit question no. 13 within first fifteen minutes.**

1. What should be the rate at which an analog signal should be sampled for conversion to digital form? 2
2. Suppose you are given a corpus of language L and a sequence of words W. How can you estimate the validity of W as a sentence of L using the idea of n-grams? 3
3. Give examples of- inflection, derivation, affixation, compound words, and non-concatenative morphological phenomenon in English. 6
4. Perform PoS tagging for the following sentence- 3  

Have you taken the medicine that the doctor prescribed you yesterday?
5. Explain with an example the phenomenon of free word order. Compare English and Hindi on this count. 2 + 2
6. (a) Give examples of sentences that are- ambiguous, and temporarily ambiguous. 2  
(b) What are some drawbacks of using *probabilistic context free grammar*. 3

7. Represent the following sentence using First Order Predicate Calculus- 2

Ravi wrote a letter to his brother from his office in the morning.

8. Identify lexical semantic relationships among the following words- 3

glass (material), live (verb), sleep, live (adjective), slip, have (auxiliary), glass (tumbler), have (eat), have (own), utensils.

9. How can feature vectors help in word sense disambiguation of the word *play* in the following sentence- 3

Ravi wrote a play that was staged on the occasion of Independence Day.

10. How do homonymy, polysemy and synonymy impact precision and recall? 3

11. Give examples of some syntactic constraints that help in resolving references. 3

12. Suggest a mechanism of estimating *fluency* of the output of translation. 3

Roll No. \_\_\_\_\_

*Detach this page and submit it within first fifteen minutes.*

13. (I) Fill in the blanks  $1 \times 10$

- (a) \_\_\_\_\_ (UTF-16 / UTF-32) is a variable length encoding.
- (b) The regular expression  $/^ [Tt]he[rs]e/$  would \_\_\_\_\_ (accept / reject) the string *these*.
- (c) The leaf symbols of a parse tree are \_\_\_\_\_ (start-symbol/terminal/non-terminal) of a context free grammar.
- (d) Application of  $\lambda$  - reduction on the expression  $\lambda(x)P(x)(A)$  results \_\_\_\_\_.
- (e) *Chair* is a \_\_\_\_\_ (homonym/hyponym/synonym) of *furniture*.
- (f) In the sentence "Ram has a truck" *has* is \_\_\_\_\_ (auxiliary/main) verb.
- (g) In information retrieval, retrieving all the relevant documents present implies highest \_\_\_\_\_ (precision/recall).
- (h) "Before Ram bought the coat, he compared the prices in several shops". In this sentence, *he* is a \_\_\_\_\_ (cataphora/anaphora).
- (i) The *fluency* of sentences produced via machine translation, can be assessed using \_\_\_\_\_ (n-grams/parallel corpora).
- (j) The sentences "Ram came to office this morning. Kamal does not like sweets." lack \_\_\_\_\_ (fluency / coherence).

(II) State whether each of the following statements are true or false-  $1 \times 10$

- (a) In Unicode BMP is one of several planes of 64 K codes.
- (b) Ambiguous natural language sentences have to be discarded during syntax analysis.
- (c) Earley's parser is a bottom-up parser.
- (d) Smoothing of n-grams is required since training corpora are never exhaustive.
- (e) "Two" and "to" are homophones.
- (f) Polysemy may lower the precision of information retrieval.
- (g) In the sentence "Kamal said that Vimal bought himself a new shirt" *himself* refers to Kamal.
- (h) The higher the entropy of a system, the higher is the predictability of its behaviour.
- (i) *Disfluency* is a phenomenon in speech.
- (j) Machine translation is more effective in restricted domains.